

The elusive ‘policy maker’

How understanding the diversity of perspectives and systems in
governments can increase the impact of scientific research

Authors: Liza Hadley^{*ab}, Slimane Ben Miled^c, Hannah E Clapham^d, Bimandra A Djaafara^{de}, Louise Dyson^f, Sebastian Funk^g, Julia R Gog^b, Hans Heesterbeek^h, Edward M Hillⁱ, Emily Howerton^j, Valerie Isham^k, Amira Kebir^{cl}, Matt Keeling^m, Stephen M Kissler^a, Justin Lessler^{nop}, Kathy Leung^{orst}, Emma McBryde^u, James M McCaw^{vw}, Chris E G Mercado^d, Denis Mollison^x, S M Thumbi^{yz}, Wirichada Pan-ngum^z, Lorenzo Pellis^z, Francisco J Pérez-Reche^z, Michael J Plank^z, Robin N Thompson^z, Cécile Tran-Kiem^z.

Author affiliations: ^aUniversity of Colorado Boulder, Colorado, USA; ^bUniversity of Cambridge, Cambridge, UK; ^cBioInformatics, bioMathematics and bioStatistics laboratory (BIMS-LR16IPT09), Pasteur Institute of Tunis, University of Tunis El Manar, Tunis, Tunisia; ^dSaw Swee Hock School of Public Health, National University of Singapore and National University Health System, Singapore; ^eOxford University Clinical Research Unit Indonesia, Faculty of Medicine, University of Indonesia, Indonesia; ^fThe Zeeman Institute for Systems Biology & Infectious Disease Epidemiology Research, School of Life Sciences and Mathematics Institute, University of Warwick, Coventry, UK; ^gCentre for Mathematical Modelling of Infectious Diseases, London School of Hygiene & Tropical Medicine, London, UK; ^hDepartment of Population Health Sciences, Utrecht University, Utrecht, The Netherlands; ⁱCivic Health Innovation Labs and Institute of Population Health, University of Liverpool, Liverpool, UK; ^jDepartment of Ecology and Evolutionary Biology, Princeton University, Princeton, NJ, USA; ^kDepartment of Statistical Science, University College London, London, UK; ^lIPEIT, University of Tunis, Montfleury, Tunisia; ^mSBIDER, Maths Institute and School of Life Sciences, University of Warwick, Coventry, UK; ⁿDepartment of Epidemiology, Gillings School of Global Public Health, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA; ^oThe Carolina Population Center, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA; ^pDepartment of Epidemiology, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA; ^qThe Hong Kong Jockey Club Global Health Institute, Hong Kong SAR, China; ^rWHO Collaborating Centre for Infectious Disease Epidemiology and Control, School of Public Health, LKS Faculty of Medicine, The University of Hong Kong, Hong Kong SAR, China; ^sLaboratory of Data Discovery for Health (D²4H), Hong Kong Science Park, Hong Kong SAR, China; ^tThe University of Hong Kong—Shenzhen Hospital, Shenzhen, China; ^uUQ Centre for Clinical Research (UQCCR), Faculty of Health, Medicine and Behavioural Sciences, The University of Queensland, Queensland, Australia; ^vSchool of Mathematics and Statistics, The University of Melbourne, Melbourne, Australia; ^wCentre for Epidemiology and Biostatistics, Melbourne School of Population and Global Health, The University of Melbourne, Melbourne, Australia; ^xDepartment of Actuarial Mathematics and Statistics, Heriot-Watt University, Edinburgh, Scotland, UK; ^yCenter for Epidemiological Modelling and Analysis, University of Nairobi, Nairobi, Kenya; ^zInstitute of Immunology and Infection Research, University of Edinburgh, Edinburgh, Scotland, UK; ^{aa}Paul G Allen School for Global Health, Washington State University, Pullman, WA, USA; ^{ab}Mahidol-Oxford Tropical Medicine Research Unit (MORU) and Department of Tropical Hygiene, Faculty of Tropical Medicine, Mahidol University, Bangkok, Thailand; ^{ac}The University of Manchester, Manchester, UK; ^{ad}School of Natural and Computing Sciences, University of Aberdeen, Aberdeen, Scotland, UK; ^{ae}Department of Twin Research and Genetic Epidemiology, School of Life Course & Population Sciences, King’s College London, London, UK; ^{af}School of Mathematics and Statistics, University of Canterbury, Christchurch, New Zealand; ^{ag}Te Pūnaha

According to Scopus, from January 1990 to February 2025, over 182,000 scientific articles referenced ‘policy makers’ or ‘decision makers’ in their title, abstract, or keywords. Some may assume that these terms refer to individuals with the ability to make decisions about government or institutional policy, informed in part by scientific information, but more than that is usually left unsaid. This audience is for example often not even defined in scientific articles. The vague abstraction of who in government can use scientific information, and in what ways, hides the true complexity and heterogeneity of real governmental systems and the roles within them. Such simplifications limit scientists’ ability to support evidence-based policy. Here we argue that it is time for scientists to move beyond viewing ‘government’ as a single, uniform, and abstract entity and instead recognise the diverse interests and roles within it. By doing so scientists will be able to contribute more effectively to government policy, both during a crisis and in the calmer times between¹.

During the COVID-19 pandemic, the interface between science and government was highly visible and intensely stressed. Scientists worldwide were abruptly thrust to the fore of a government decision-making process that was hectic, complex, and for some unfamiliar. The authors of this article are scientists who were involved in the pandemic response in 14 different countries representing every region of the globe. Through these experiences and other crisis and non-crisis work since, we gained important insights into how scientists might better understand the complex structure of government decision-making and the diversity of actors involved². Building on these experiences and a qualitative study of modelling-to-policy interactions [1], we present a general framework to help scientists better understand this diversity of actors

¹ Note that ‘science’ and ‘government’ are not always exclusive categories and are not intended as such. We consider a ‘scientist’ to be any individual who identifies primarily or wholly as a scientist, such as those who lead or perform scientific investigations. For the purposes of this article, we also define ‘policy’ to be a law, regulation, procedure, administrative action, incentive, or voluntary practice of governments and other institutions. This article focuses on national government policy.

² This paper arose from a one-month invited program held at the Isaac Newton Institute of Mathematical Sciences in Cambridge, UK in August 2024. The authors are from 14 countries, representing all six World Health Organization geographic regions. All authors have experience in working with policy responses as epidemiological modellers at the national or international scale, including supporting their respective governments in the COVID-19 pandemic but many also have experience in numerous

outbreaks and other health events prior and since.

and interests as they work to effectively support evidence-based policy, both in peacetime and times of crisis.

Who is the ‘policy maker’?

Governments are multi-faceted, complex systems made up of heterogeneous actors each with different objectives, constraints, and timelines. Though it is impossible to capture this complexity with a simple taxonomy, we find it useful to group actors into three categories based on their relationship to the strategy (i.e., broad goals and overall architecture) and tactics (i.e., specific actions and implementation) of government policies:

- **Policy specialists**, who have working-level (i.e., tactical) responsibility for the specific topic area;
- **Policy generalists**, who are focused on broader, strategic activities and may not have specialist knowledge in the specific area;
- **Decision makers**, who are senior public servants and politicians who set overall strategic policy goals and have statutory decision-making powers.

For example, the US CDC’s subject matter experts or Kenya’s public health officers would be ‘policy specialists’; a Chief Epidemiologist for a US state or a Kenyan department Director General would be a ‘policy generalist’; and elected politicians or appointed ministers would be ‘decision makers’. The distinctions between these groups are not black and white; individuals move between these roles and there is considerable variation within groups³. Nonetheless, we argue that these are useful categorisations for scientists to consider because the objectives, experience, and remit of each of these actors differ. To facilitate effective dialogue, the nature of the relationship between scientists and each of these actors should also differ.

³ Individuals may hold different roles, and may move between policy and decision making roles, or even hold roles with overlapping responsibilities (e.g., when a legislation or regulation delegates a certain decision-making capability to a public servant). Nonetheless, at any given time or in any given context, an individual can be considered to hold a particular role as we

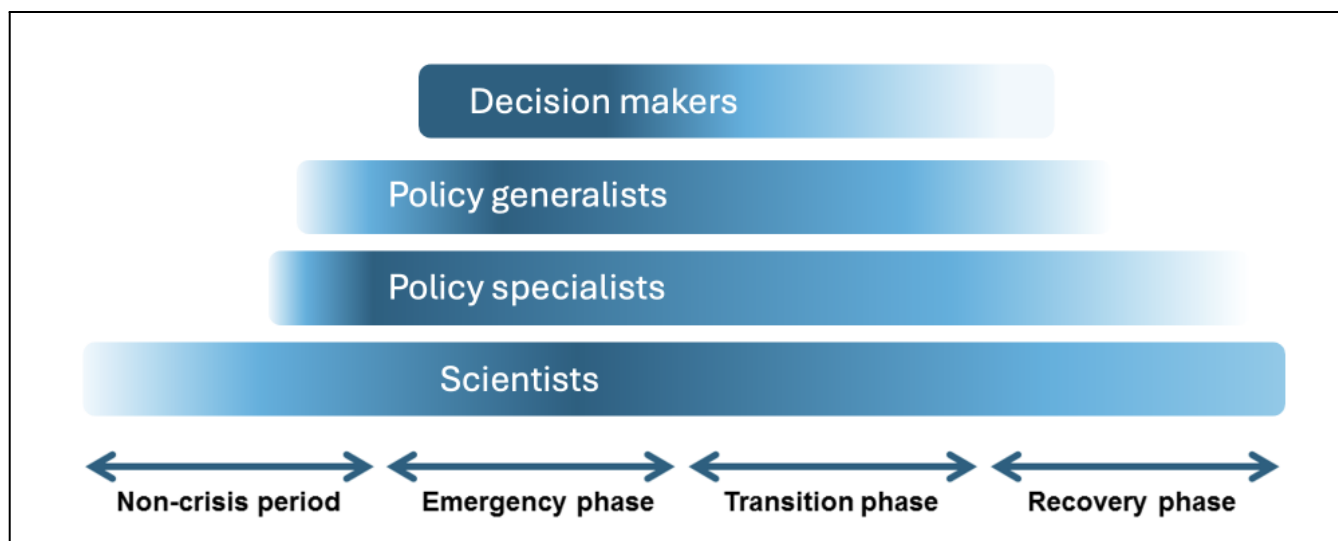


Figure 1: Different actors in government will have a different level of involvement in a health crisis or other emergency depending on the timeline of the threat (horizontal axis represents time). Scientists are typically working on their respective topic at all times, including non-crisis periods. Policy specialists and generalists may be involved on a broad topic in non-crisis periods (e.g. on general preparedness and planning for future emergencies), but may become increasingly engaged with a specific issue as it poses concern or requires monitoring. High-level decision makers are likely to be directly engaged only when the issue poses an acute high-level threat or emergency, with little to no involvement with scientists and researchers in recovery and non-crisis periods.

For any particular science-policy issue, the level of government actors involved will likely depend on the event's acute impact and where it sits in the scale of political importance (Figure 1). Much of the time, detailed exchanges may only be between scientists and policy specialists. When the threat level is higher, scientists may be requested to engage directly with generalists and decision makers. This can occur either organically or through the explicit creation or reassignment of groups that include government actors with a mix of roles in response to the crisis. Two examples are illustrative: (1) In mainland China prior to the COVID-19 pandemic, academic scientists worked with infectious disease policy specialists at the Chinese Center for Disease Control and Prevention (China CDC). However, a separate deputy-ministerial-level agency called the National Administration of Disease Control and Prevention was convened in 2021 in response to the COVID-19 pandemic. In this agency, academic scientists had direct contact with policy generalists. (2) Similarly in Tunisia, government institutions like the National Observatory of New and Emerging Diseases focus on long-term surveillance and preparedness during non-crisis periods, facilitating interaction between academic scientists and policy specialists. During the COVID-19 pandemic, distinct rapid-response bodies like the National Coronavirus Response Authorities were activated to coordinate

efforts across regions and sectors; these entailed more direct interaction between academic scientists and policy generalists. Post pandemic, responsibility for communicable disease management has largely returned to the expert specialist bodies in China, Tunisia, and many other countries.

There are potential risks/challenges if policy specialists are bypassed as in the examples above. For instance, policy generalists and decision makers may be inexperienced in understanding the language and scientific principles of a field, leading to misinterpretation of evidence and recommendations. Likewise, the experience and knowledge they gain over the course of a crisis may be lost once high-level generalists return to other priorities, hence lessons may not become embedded in post-crisis business-as-usual practices.

Building and maintaining science-policy relationships

Communications between scientists and government actors need to be tailored to the attention, level of detail, and turnaround time required by different government actors. Relationships with policy specialists will often create the personal ‘who am I going to call’ networks and crucial trust in information sharing. Sustained relationships with policy specialists not only deliver benefits during crises, but support non-crisis activities, through development of in-house scientific capacity, more targeted foundational research, and more rapid adoption of scientific advances into policy [1–4]. In contrast, relationships with policy generalists and decision makers typically centre on broad strategic relevance and also enable familiarity with particular types of scientific outputs. These actors may engage with specific scientific subfields less frequently or with less technical depth, so scientists must adjust their communication accordingly. In Table 1 below, we provide examples to illustrate how relationships and routes of communication may differ when working with policy specialists, policy generalists, or decision makers. Level of detail, timing, and objective varies.

	Example roles	Case studies from communicable disease epidemiology
Policy specialists	Health officer, public health agency subject expert.	In Indonesia, long-term collaborations between academic epidemiological modellers and policy specialists in Dengue and Malaria Control enabled direct requests to come through to scientists, with direct interactions with each disease sub-directorate in the Ministry of Health. Engagements included discussions, co-creation of questions, and two-way feedback through regular calls and physical meetings. Similar routes of communication are seen in Thailand, where modellers could engage with specialist policy during the pandemic through existing (non-crisis) collaboration within the Emerging and Re-Emerging Infection Disease program, part of the Ministry of Science and Technology.
Policy generalists	General public sector policy maker, department Director Generals, chief scientific advisors, chief medical officers.	In China during the COVID-19 pandemic, academic modellers supported senior actors in the China CDC (policy generalists), where they were asked for ad hoc reporting and to communicate directly with the CDC when discussions were needed. Most reports were concise (<1000 characters) or in powerpoint format. During the Omicron wave, these modellers were also requested to visit the CDC to provide real-time reporting. In New Zealand, this short line of communication to decision makers similarly needed a brief, concise summary of results, e.g. a simple graph or soundbite, or a one-sentence response to a question. As a second example, the World Health Organization regional offices for South-East Asia (SEARO) and the Western Pacific (WPRO) facilitated links with policy generalists in countries that had limited scientific capacity, connecting academic modellers from Australia directly to Chief Medical Officers in Malaysia, Indonesia, Fiji, and French Polynesia.
Decision makers	Prime minister or president, governors, ministers.	In Singapore, COVID-19 results for decision makers were primarily distilled into key powerpoint slides providing the 'big picture' findings for policy generalists (in contrast to working with policy specialists where the academic modellers could discuss methods, model structure, parametrisation, and future data needs directly). A few academic modellers in Europe and Australia were seconded to their national governments' Cabinet (the senior decision making level) or sub-committee during COVID-19, where they played a primarily interpretative role. There were often faster turnaround times than working with policy specialists and most of the work in this example involved interpreting and translating a broad literature. Here it was clear that communicable disease epidemiology formed one part of a much larger picture. In South Africa, modellers played a similar role through appointment to the Ministerial Advisory Committee on COVID-19, where they worked in an

		interdisciplinary group to provide policy advice to the Minister of Health in response to questions on a wide variety of topics.
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Table 1: Who do scientists interact with in government? Examples of how relationships may differ when interacting with different government actors, such as policy specialists vs policy generalists vs decision makers.

Going deeper, relationships between science and government require both tangible ‘hardware’ such as data sharing agreements, memorandums of understanding, and advisory group membership, as well as intangible ‘software’ such as mutual values, norms, relationships, and trust [5]. The hardware underpinning these relationships is particularly important to policy specialists, who need legal support for sharing information and often work within strict regulatory frameworks. It is critical to maintain both hardware and software during non-crisis periods to enable rapid scale-up when new threats emerge. For example, several COVID-19 modelling groups supported by government funding agencies in Australia benefited from existing trusted relationships with policy generalists in the Commonwealth and State Health departments (established software), but, without critical agreements in place, data-sharing was slow and structural capacity to build consensus through open science was insufficient (lacking hardware). The crisis catalysed change, including commitment to establish an Australian Centre for Disease Control (a home for policy specialists with established scientific relationships) [6], but many of these changes only materialised post-crisis. It is critical that scientists approach the establishment of such relationships or agreements with government with an awareness that decision makers and other elected government individuals may view the response to topics in the public eye through a political lens, while for scientists the independence from such matters is paramount.

Another important consideration is personnel turnover and shifting attention in government, which makes one-to-one relationships fragile. As highlighted in Figure 1, different levels of actors may be focused on a particular topic at different times. This motivates an approach to engagement which is constantly establishing and renewing relationships between individuals, institutions, agencies and groups. Examples of facilitating such activities through hardware include: systematic documentation of interactions and

relationship-building processes; academic representation on committees; active involvement in general/specialist policy forums; dual appointments/secondments; and co-funded research by governments where possible to create a sense of ownership and independence from external/foreign funding where relevant.

What else can scientists do?

Appropriate and nuanced relationships between scientists and different government actors are the most important ingredients if science is to contribute to effective preparedness and response. Training (for both scientists and government actors) is important for getting the most out of these relationships due to the differences in primary responsibilities and focus. Much of the time this ‘training’ can happen informally as relationships are built and maintained. We suggest three specific actions for external scientists seeking to actively engage in policy:

1. It is crucial to learn (and adopt) the *language* and communication culture of the different government actors we are working with. While policy specialists are often comfortable with technical language, there are different language needs for decision makers. At times, words can have specific meanings or regulatory implications, such as the term ‘activate’ describing certain regulatory enactment in the US, and the term ‘probable’ relating to a government framework of language for discussing probabilities in the UK [7]. Points that seem elementary to those working in the field may not be well known to policy generalists and decision makers with different expertise, and should be explicitly stated. Beyond vocabulary, preferences may differ around formats and timescales for analyses [8,9]. The different languages used by different government actors may require, for example, ‘stepped’ reports that present a high-level summary for decision makers, followed immediately by a short report for policy generalists, and supplementary detailed scientific analysis for policy specialists. The communication of uncertainty, though critical, is often an area where translation between the language of scientists and the language of policy generalists and

decision makers breaks down [10]. How to best overcome this challenge has been the source of much research and debate, and scientists should familiarise themselves with this background and tailor their communications accordingly [11].

2. Communicating as a central coherent voice can also aid the uptake of science. In the same way that governments can be difficult for scientists to navigate, the science community can be difficult for government actors to navigate too. In countries fortunate enough to have multiple experts engaged on a given topic, we argue that a diversity of perspectives should be actively solicited, discussed, challenged, and refined by the relevant scientific groups [12,13]. Learned societies can provide one option for aiding solicitation and enabling a forum for new scientists who want to contribute [14]. However, as far as possible, views should be synthesised and communicated to decision makers, and to a lesser extent policy generalists, as a coherent body of evidence. In some contexts, it may be preferable for this distillation and translation of evidence to be performed within government by policy specialists, by scientists and policy specialists collaboratively, or by independent translators/intermediaries.
3. Lastly, decision science could also provide a pathway for translating scientific endeavour to evidentiary support for actions. Doing so would require a subset of researchers to follow the structured approaches of decision science and support government actors (particularly policy generalists) in framing their problems in ways that these techniques can be applied [15–17]. By acknowledging that the decision process must also account for social, political, and organisational factors, the contribution of science to decision making can be better understood.

Concluding remarks

We have sought to illustrate how relationships between science and government may differ depending on the type of government actor involved (Figure 1, Table 1). If scientists do not acknowledge this diversity of

actors in government decision-making, and oversimplify their understanding of the decision-making processes, it may lead to miscommunication and a reduced impact of scientific evidence in public policy. Conversely, a greater understanding and explicit consideration of the diversity of actors within government can support more impactful contributions of science to policy in emergency and routine decision-making.

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Author contributions

All authors took part in discussions and contributed to the writing of the manuscript. LH coordinated discussions throughout and led the writing and editing of the manuscript. All authors provided feedback on the manuscript and approved the final version for publication.

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